REVEALING "MAFIA INC."? FINANCIAL CRISIS, ORGANIZED CRIME, AND THE BIRTH OF NEW ENTERPRISES

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Abstract—We study the investment of organized crime in the legal economy. By using the shock induced on the Italian credit market by the 2007 subprime mortgage crisis, we document how provinces with a high organized crime presence have been affected less by the crisis in terms of the establishment of new enterprises than provinces with a lower criminal infiltration. We provide evidence that the lower impact of the crisis is consistent with the presence of investments by organized crime in the legal economy. We corroborate this interpretation by comparing our results with the characterization made by the judicial authority of such investments.

I. Introduction

THE presence of criminal organizations has been found to be largely detrimental for economic prosperity and development (Peri, 2004; Pinotti, 2015). Although organized crime usually operates through the use of violence, killings, and corruption to obtain power and territorial control, it also represents an important economic actor in many countries. According to the most recent estimates produced by the United Nations Office on Drugs and Crime (UNODC), the worldwide proceeds of organized crime in 2009 accounted for US$2.1 trillion, with US$1.6 trillion reinvested in the legal economy (UNODC, 2011).

In this work, we study organized crime's investment in the legal economy.[1] This investment is important in different dimensions. On the one hand, it is one of the main channels that criminal organizations use to launder money (Ardizzi et al., 2014) and make profits (UNICRI, 2016). On the other hand, it might be an important tool for organized crime to raise forms of social consensus in a segment of the population by acting as an alternative provider of capital and jobs (Gambetta & Reuter, 1995).[2]

We focus on Italy, a country that has been plagued by several criminal organizations since the nineteenth century, and we study the effect of the presence of organized crime on the establishment of new enterprises. We exploit the shock on the Italian credit market induced by the 2007 subprime mortgage crisis and compare in a difference-in-differences (DiD) setting the number of newly established enterprises in the pre- versus postcrisis period between provinces with and without mafia presence. We define mafia presence according to the Transcrime Mafia Index (TMI).[3]

Italy features three key elements underlying our identification strategy that allow us to detect the mafia investment in the legal economy. First, National Law 580/1993 requires every enterprise operating in Italy to be registered in the Registry of Enterprises. Therefore, the registry contains both legal enterprises and enterprises with hidden connections to organized crime. Second, the 2007 subprime mortgage crisis induced a credit contraction that was exogenous and hit areas with and without mafia presence with similar intensity. The crisis originated in the United States and rapidly propagated, without anticipation effects, to the Italian legal credit market, inducing a sizable contraction in terms of credit granted to the entrepreneurial sector. We show that provinces with higher or lower levels of mafia presence experienced similar contractions in the years immediately after the onset of the crisis. Third, the crisis had a limited impact on mafia sources of capital and profits. UNODC (2011) reports no impact, or even some positive impact, of the crisis on mafia profits. As confirmed by the analysis of the main source of profits for Italian organized crime, the market of illegal drugs, organized crime mainly raises capital in markets barely affected by the economic downturn.

This framework of mandatory enterprise registration, exogenous and homogeneous credit contraction across the country, and nondecreasing organized crime financial resources makes mafia infiltration into the economy empirically detectable. Indeed, the onset of the subprime mortgage crisis depressed the supply of legal credit available for entrepreneurs similarly in provinces with and without a mafia presence. Consequently, the decrease in the number of established enterprises observed in the registry should be similar across provinces with and without mafia. However, in provinces with a mafia presence, the number of established enterprises should decrease less under the hypothesis that

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[1]For the remainder of the study, we use the expressions *organized crime, criminal organizations*, and *mafia* as synonyms.

[2]Social consensus is one of the fundamental milestones of the empowerment of criminal organizations. For instance, the Sicilian Mafia in Italy rose to power in the nineteenth century as a movement fulfilling the need of the population to protect its land from predatory attacks (Bandiera, 2003). Similar cases have arisen worldwide (e.g., terrorist organizations and drug cartels).

[3]The TMI represents the most comprehensive definition of mafia presence. Here, as in the remainder of the paper, we use the expression with (without) mafia presence to mean with higher (lower) mafia presence.

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mafia invests in the legal economy. We test this hypothesis in our DiD setting.

The baseline analysis shows that provinces with higher mafia infiltration experience a less severe drop—about 4%—in the number of new enterprises established in the postcrisis period. This effect corresponds to a provincial average of 241 enterprises established every year in the postcrisis period due to mafia investment in the legal economy. Results are robust to a large number of tests for sample selection criteria, exclusion or inclusion of different sets of control variables, and possible residual endogeneity not captured by the DiD setting. More important, results do not depend on how mafia presence is defined.

We run a set of validation, placebo, and falsification tests to corroborate the interpretation of the baseline results as evidence of mafia investment in the legal economy. We take advantage of the specific characteristics of the mafia investment in the legal economy revealed by the Italian judicial authority. Judicial evidence shows that mafia prefers to operate in the construction sector or under the legal form of a limited company. At the same time, the mafia invests few resources in sectors characterized by high levels of innovation, such as the professional, scientific, and technical sector (R&D). Our interpretation of the baseline results implies a sizable postcrisis effect of mafia presence in sectors or legal forms with a traditional presence of mafia. Conversely, there should be no or a reduced effect of mafia presence for sectors rarely preferred by mafia. The analysis confirms judicial evidence displaying a large postcrisis effect of mafia presence, specifically, less-severe drops in new enterprises in the construction sector and in enterprises with the legal form of a limited company; no effect arises for sectors avoided by mafia such as the R&D.

Through additional falsification tests, we rule out the possibility that postcrisis changes in important determinants of the establishment of new enterprises could drive the results. These tests confirm that the baseline results are not driven by a different impact of the crisis on the demand for goods, a different degree of openness to the global market, or by differential postcrisis changes in the accessibility of economic resources such as public procurement or EU funds between provinces with and without mafia presence.

As a final step, we extend the analysis to closed enterprises and to the stock of operating (registered) enterprises to provide an overview of the overall impact of mafia presence on the local entrepreneurial activity.[4] The analysis points to a higher, but statistically insignificant, postcrisis increase in closures in provinces with mafia than in provinces without mafia. Practices such as usury and racketeering likely exacerbate entrepreneurs' difficulties in dealing with the consequences of the crisis. The effect of mafia presence on closed enterprises is smaller in size than the effect on new enterprises, and this difference results in a positive net effect on registered enterprises.

This work represents the first attempt to reveal the actual investment of organized crime in the legal economy through evidence of its impact on the degree of local entrepreneurial activity. In particular, our study is the first to highlight the existence of positive-in-sign effects of mafia territorial presence and provide an explanation for the persistence of such presence. In fact, although it is hard to empirically identify the specific scope underlying this investment, anecdotal and juridical evidence suggests that through the investment in the legal economy, organized crime is likely to accomplish tasks such as money laundering, generating profits, and raising forms of social consensus in the local population. The latter is often responsible for the ineffectiveness of policies targeted at reducing leniency toward organized crime. We also shed light on different aspects of mafia investment in the legal economy. Specifically, by confirming the existing qualitative evidence of mafia preferences when it comes to investing money in terms of economic sectors of activity and adopted legal forms, our research pinpoints where monitoring and vigilance should be improved.

The remainder of the paper is structured as follows. Section II briefly describes Italian organized crime. Section III introduces the data and the identification strategy underlying the analysis. Section IV discusses the main results of the paper. Section V shows the validation, placebo, and falsification tests. In section VI, we extend the analysis to closed and registered enterprises. Section VII concludes the work.

II. Organized Crime in Italy

Italy is characterized by the existence of a complex and well-established system of criminal activities managed by mafia-type organizations, which is exceptionally heterogeneous across the country. The Sicilian Mafia (Cosa Nostra) rose in the nineteenth century in response to the population's demand for private protection in a context of weak institutions and valuable natural resources (Gambetta, 1993; Bandiera, 2003; Buonanno et al., 2015; Dimico, Isopi, & Olsson, 2017). The expansion of Cosa Nostra in Sicily was contemporaneous with the birth of similar criminal organizations, such as the Camorra in the Campania region and the 'Ndrangheta in the Calabria region.

After initial immobility, criminal organizations started to expand to the more productive and profitable central-northern regions. Factors such as the massive migration flows from southern Italy triggered by the "Italian economic miracle" in the late 1950s until the early 1970s, and the *confino* law, which in the 1960s and 1970s imposed resettlement to different provinces on individuals likely to be involved in mafia-type criminal activities, pushed this expansion (Buonanno & Pazzona, 2014).

[4]The baseline analysis focuses on new enterprises to isolate mafia capital inflows to the legal economy as forms of investments and to mitigate the effect of other mafia interference in local entrepreneurial activity such as usury. Unioncamere (2014), Sos Impresa (2012), and Dalla Pellegrina (2008) show that new enterprises are less affected by illegal mafia lending practices and that these practices more commonly affect operating enterprises.

The effect of mafia presence is the topic of several empirical studies. Mafia expansion and presence is found to deteriorate political competition and to affect electoral outcomes (Daniele & Geys, 2015; Daniele & Dipoppa, 2017; De Feo & De Luca, 2017; Alesina, Piccolo, & Pinotti, 2019; Acemoglu, De Feo, & De Luca, 2020; Di Cataldo & Mastrorocco, forthcoming). Mafia presence also generates sizable losses in terms of economic resources such as local GDP or productivity (Peri, 2004; Albanese & Marinelli, 2013; Pinotti, 2015). Economic losses result from the mafia's ability to control local entrepreneurial activity and obtain monopolistic power by violent and intimidating behavior (Reuter, Piccolo, & Pinotti, 1982; Reuter, 1987; Falcone, 1991; Reuter, 1993; Savona and Berlusconi, 2015; Calamunci & Drago, 2020). Moreover, in regions characterized by high levels of criminal activity, the cost of credit tends to increase (Bonaccorsi di Patti, 2009), foreign direct investments are lower (Daniele & Marani, 2011), and misallocation of public funds in the form of business subsidies occurs (Barone & Narciso, 2015; Daniele & Dipoppa, 2019).

This study complements the existing literature on organized crime by shedding light on possible positive-in-sign spillover effects of mafia presence due to its investment in the local legal economy. The understanding of possible positive-in-sign effects of mafia presence is important to understand how organized crime obtains social consensus among some of the population (Murphy & Rossi, 2020).

III. Revealing the Mafia Investment

In this section, we introduce the data used in the analysis and the elements underlying the identification strategy to unveil the mafia investment in the legal economy. We then describe the implementation of the identification strategy and discuss its main assumptions.

A. Data

We assemble a new panel data set containing information on established enterprises, mafia presence, and a set of control variables for determinants of the establishment of new enterprises.[5] Each panel observation is at the province-year level. In the baseline analysis, we exclude from the sample provinces in the three regions where mafia originated, namely, Sicily, Campania, and Calabria, as these regions considerably differ in structural and economic terms from the rest of the country.[6] As a result of this sample selection, the data set covers 84 Italian provinces for 2003 to 2013.[7]

The main dependent variable of interest is the number of new enterprises established at the province-year level. We exploit National Law 580/1993, which requires every Italian enterprise to register its activity on the Registry of Enterprises. This registration is mandatory for all enterprises that operate nationwide with the exception of professions such as lawyers or architects. We collect data on the number of new registrations in the Registry of Enterprises at the provincial level from the Italian National Institute of Statistics (ISTAT).

We adopt the Transcrime Mafia Index (TMI) as the baseline measure for mafia presence.[8] The TMI constitutes the most comprehensive definition of mafia presence and includes features related to both territorial infiltration in terms of military occupation and the mafia's capacity to provide illegal goods and services. The TMI is a composite index at the provincial level and is constructed using records of those convicted of being in a mafia-type organization as defined in Law 646 and art.416-bis, murders committed by mafia members, city councils dissolved because of mafia infiltration, and assets seized due to organized crime.

We define mafia presence as an indicator variable *Mafia* based on the tertiles of the TMI distribution. We opt for the use of tertiles of the TMI distribution to obtain a clearer contrast between the group of provinces with and without mafia presence. However, as there is no univocal way to define mafia presence, appendix A.4 reports a detailed sensitivity analysis based on different measures and definitions for mafia presence. The analysis reassures that none of the baseline results depend on the measure or definition we use for mafia presence.

The indicator *Mafia* takes the value of 1 for provinces within the last tertile of the TMI distribution and 0 for provinces within the first or second tertile of the distribution.[9] Figure A.9 in the appendix validates the choice of combining the first and second tertiles of the TMI. Indeed, the first and second tertiles of the TMI distribution shape the establishment of new enterprises in a similar way. Figure 1 shows the territorial distribution of the TMI according to the tertiles of its distribution.

Our data also include a set of control variables for determinants of the establishment of new enterprises potentially correlated with local mafia presence. Control variables capture the structure of the local credit market as well as the characteristics of the economic, institutional, and social environments.

Big banks, which are able to deal with higher and riskier credit costs, represent a good approximation for credit market conditions. The characteristics of the local credit market might shape entrepreneurial activity as well as the perpetration of criminal activities such as usury or money laundering

[5] Appendix A.1 contains a detailed description of all data sources used in this study.

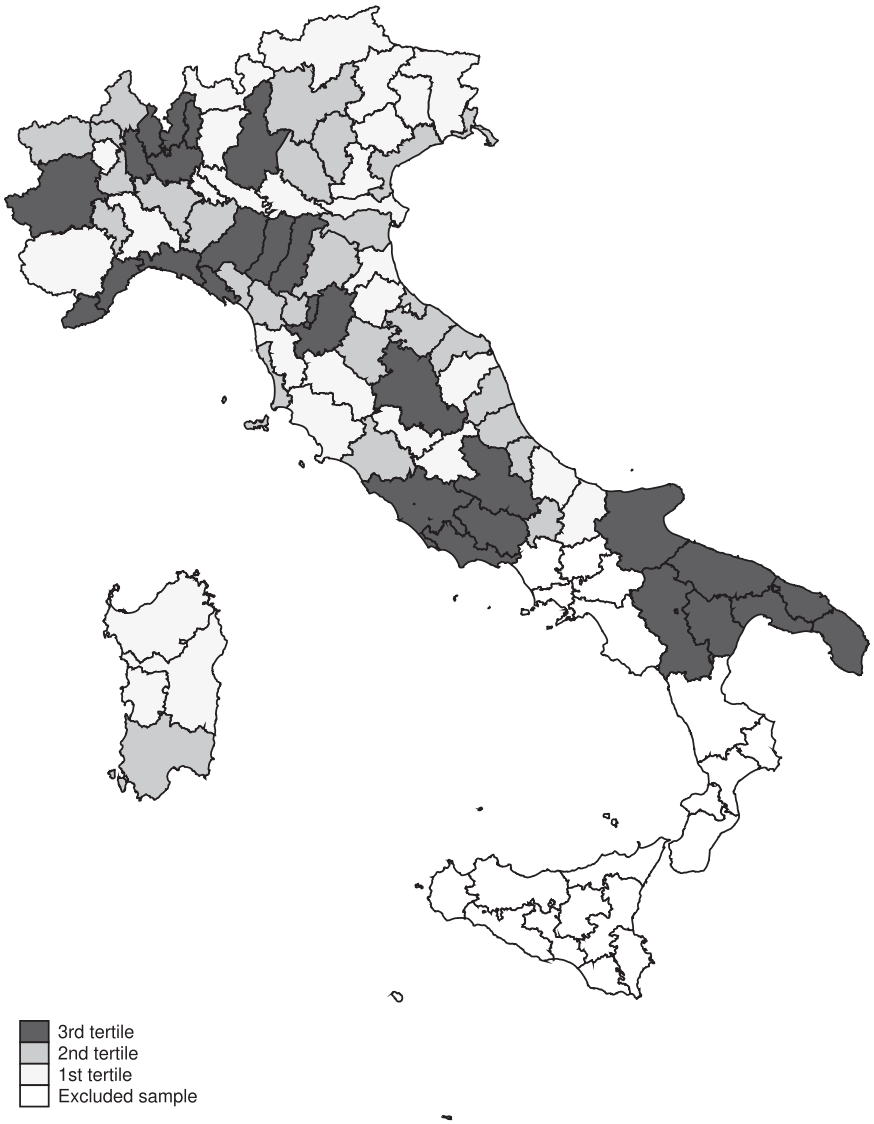
[6] Appendix A.5 provides a robustness test for our baseline analysis based on the sample containing all Italian regions.

[7] We start the analysis in 2003 because of pre-2003 data unavailability for the control variables described below.

[8] The TMI was developed by Calderoni (2011). In the baseline analysis, we rely on the version of the index provided in the report *Mafia Investments*, funded by the Italian Ministry of Internal Affairs and the European Union (Ministry of the Interior, 2013). Appendix A.2 provides a detailed description of the TMI.

[9] We compute the tertiles of the TMI distribution after excluding the three regions not part of the main analysis: Sicily, Campania, and Calabria. In the remainder of the paper, we label provinces with mafia versus provinces without mafia according to this definition.

FIGURE 1.—GEOGRAPHICAL DISTRIBUTION OF THE TMI



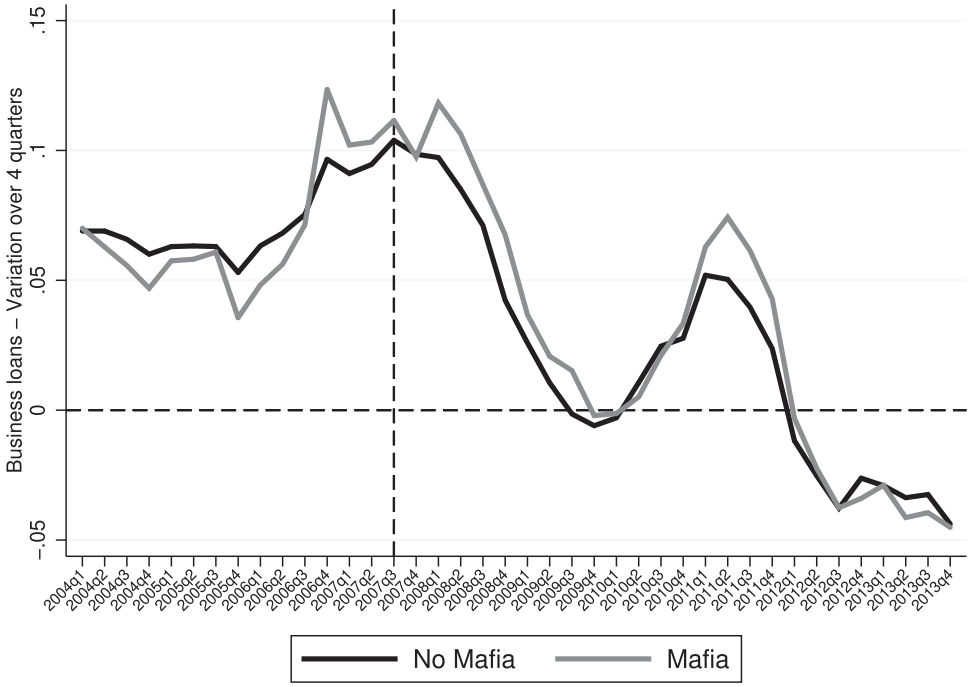
This figure shows the geographical distribution of the Transcrime Mafia Index (TMI). The map shows the tertiles of the TMI distribution across the Italian provinces. The more intense the shading of a province, the higher the tertile to which it belongs. White provinces are those excluded from the sample.

(Dalla Pellegrina, 2008). We gather data on the fraction of big banks (banks whose total value of traded funds is greater than EUR 26 billion) over the total number of banks to proxy the characteristics of the local credit market.

For the economic environment, we calculate the share of self-employment over total employment in a certain province to control for the local degree of entrepreneurship. Self-employment positively correlates with business creation (Faggio & Silva, 2014). As some economic sectors, such as tourism and waste treatment, are particularly at risk of criminal infiltration, we focus on these two sectors as markers of provincial attractiveness for mafia (Ministry of the Interior, 2013; Savona, 2015). We measure the relevance of the tourism and the waste treatment sectors with an index of attractiveness of tourism-related consumption and the per-capita quantity of produced waste.

We proxy the quality of the institutional and social environment with control variables for the efficiency of the judicial system, urban population, newspaper circulation, and blood donations. The efficiency of the judicial system is measured as the provincial average duration in days of a bankruptcy trial. The efficiency of the judicial system stands out as one important factor that positively affects the level of entrepreneurial activity (Chemin, 2009) and negatively affects crime (Blanco, 2012). We compute the fraction of population living in urban areas at the provincial level as both new-enterprises formation (Delfmann et al., 2014) and crime (Christens & Speer, 2005) are affected by the urban-living population. We also collect data on newspaper circulation within each province and the regional average number of blood donations to account for the level of human and social capital, respectively (Acs & Armington, 2004;

FIGURE 2.—CHANGE IN THE SUPPLY OF LEGAL CREDIT TO THE ENTREPRENEURIAL SYSTEM



This figure shows the change over four quarters in the amount of credit granted to the Italian entrepreneurial system. The comparison is between provinces within the first two tertiles of the TMI distribution (No Mafia, black line) and those belonging to the third tertile (Mafia, gray line). The black vertical dashed line indicates the quarter of the year in which the financial turmoil at the root of the subprime mortgage crisis started.

Buonanno, Montolio, & Vanin, 2009; Nieto & González-Álvarez, 2016).[10] Table A.4 in the appendix provides summary statistics for our sample.

B. *The Identification Strategy*

Our identification strategy relies on three elements: (a) the registration in the Registry of Enterprises required of each enterprise operating in Italy; (b) the exogeneity and geographical homogeneity of the credit contraction induced by the 2007 subprime mortgage crisis; and (c) a lower impact of the crisis on mafia sources of capital and profits than the impact observed in the legal credit market.

The first element of the identification strategy is the mandatory registration of Italian enterprises. This registration allows using information about licit markets to infer insights about illicit activities (Zitzewitz, 2012). Illicit trafficking and activities often interface (and interfere) with legal markets. This relation leaves detectable traces as it is subjected to the laws and rules governing legal activities. As a result of the mandatory registration in the Registry of Enterprises, the registry contains information on the universe of enterprises operating in the country, among which is a subgroup with hidden connections to criminal organizations.

The second element of the identification strategy is the shock induced by the 2007 subprime mortgage crisis. Ideally, in order to compare possible consequences induced by the shock in provinces with different levels of mafia presence, we would need a sizable and exogenous credit shock. Moreover, the shock would need to similarly hit provinces with and without mafia presence. The subprime mortgage crisis fulfills these two requirements.

The shock induced by the subprime mortgage crisis was not anticipated by the Italian credit market as it started in the United States in 2007 and rapidly spread to the rest of the world. This lack of anticipation is confirmed by the difference between interbank interest rates, namely, the LIBOR for the United States and EURIBOR for the European Union, and the interest rate on the equivalent index swap rate. This difference is a common measure of liquidity of the banking system and the counterparty risk. The measure reports negligible differences up to the second half of 2007. In August 2007, the difference suddenly starts to rapidly increase to its peak in the second half of 2008 with the breakup of Lehman Brothers. The loss of confidence among financial institutions was directly reflected in their supply of credit to borrowers. According to the diffusion index created by the European Central Bank (ECB) and the Bank Lending Survey (BLS), a measure capturing the evolution of credit standards, Italian loan officers started to report a tightening of the lending standards almost contemporaneous with the beginning of the financial turmoil in the United States in 2007.[11]

The tightening of lending standards affected the amount of credit available for Italian entrepreneurs. Figure 2 plots the rate of change of credit supplied to the entrepreneurial

[10] Data about blood donations are unavailable at the provincial level.

[11] Appendix A.3 provides the description and the analysis of the spread and the diffusion index.

system based on data by the Bank of Italy. Each point in the figure represents the variation in the credit supplied to the entrepreneurial system with respect to the same quarter of the previous year. No anticipation effect appears in the precrisis period. Credit grows at a constant rate in the initial 2004–2005 period. In 2006, the provision of credit shows increasing (positive) growth rates. The (relative) increasing trend in credit granted to the entrepreneurial system suddenly stops in 2007 and becomes almost flat before starting a decreasing trend.

Figure 2 also shows that the credit contraction has similar impacts on provinces with and without mafia presence. In table A.5, we replicate the analysis through a formal regression model. We aggregate the rate of change of credit supplied to the entrepreneurial system at the province-year level. We regress the rate of change of credit on the interaction term between *Mafia* and *Crisis*, an indicator variable for 2007 to 2013. Then we replicate the analysis by focusing on the interaction terms between *Mafia* and indicator variables for each year in the 2004–2013 period. Results in both specifications highlight that none of the point estimates is statistically significant. In the precrisis period, provinces with and without mafia appear on similar trends in terms of credit provided to the entrepreneurial system. Also in the postcrisis period, provinces with and without mafia remain on similar trends, which confirms the homogeneity of the impact of the subprime mortgage crisis.

The third element of the identification strategy requires the subprime mortgage crisis to affect mafia profits and capital with a lower intensity with respect to the hit on the legal credit market. The report UNODC (2011) indicates that in the postcrisis years 2007 to 2009, the revenues generated by organized crime in Italy through illicit traffic remained not only largely stable but even showed a small increase. This increase suggests that Italian organized crime might have benefited from the crisis. We empirically test this assumption with an analysis of the Italian illegal drugs market. Organised Crime Portfolio (2015) and Savona and Berlusconi (2015) show that drug dealing generates the highest profits for organized crime in Italy. According to Transcrime, drug dealing generates EUR 7.7 billion per year, an amount almost double that of criminal organizations' second source of revenue, racketeering.

Figure 3 investigates the possible impact of the crisis on the Italian market for illegal drugs in the period of interest for this study. We focus on data by the European Monitoring Centre for Drugs and Drug Addiction on four drugs—amphetamines, cocaine, heroin, and cannabis—and four market indicators—the total number of offenses for drug use (figure 3a), the total number of seizures (figure 3b), the total quantity of drugs seized (figure 3c), and the average price (figure 3d). Illegal drug demand (and consumption) is stable over time. Figure 3a shows that the number of offenses related to drug use has been almost constant since 2007. The exception is cannabis offenses, which saw a (slight) decrease. On the supply side of the market, both the volume (see figures 3b and 3c) and prices (see figure 3d) remain almost stable during the entire period, except for an increase in the quantity of cannabis seized and a modest drop in the price of cocaine and heroin.[12]

To summarize, the context of mandatory enterprise registration, exogenous and homogeneous credit contraction across provinces with and without mafia, and illegal capital invested in the legal economy makes mafia-connected enterprises more detectable given that provinces with stronger mafia presence should experience a less-severe postcrisis drop in the number of new enterprises registered every year.

C. *The Difference-in-Differences Estimator*

We implement the identification strategy described in section IIIB in a DiD setting based on two alternative specifications: a pooled DiD setting and an event study setup.

In the first specification, we pool postcrisis years and compare the number of new enterprises (a) before and after the outbreak of the subprime mortgage crisis and (b) between provinces with and without mafia. The DiD baseline equation of interest in this case is

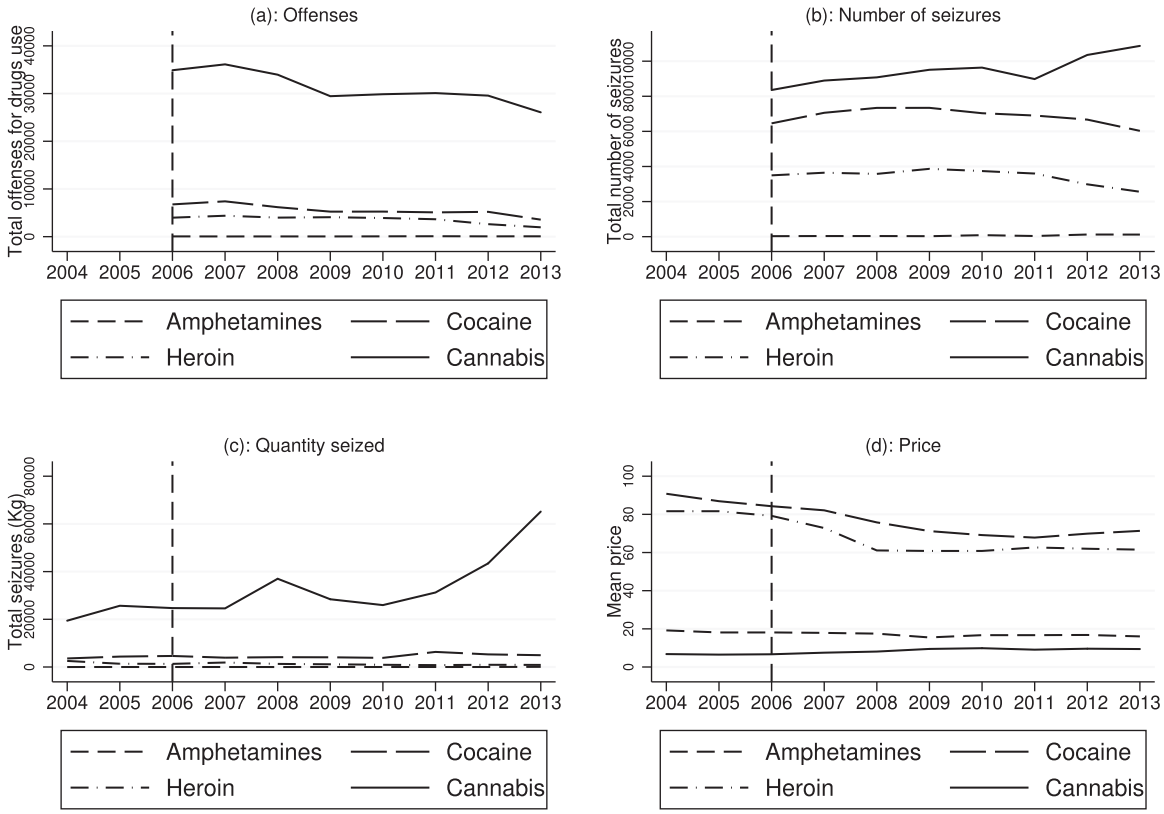
$$NewEnterprises_{i,t} = \beta_0 + \beta_1 Mafia_i \times Crisis_t + \mathbf{X}'_{i,t}\boldsymbol{\beta} + \alpha_i + \gamma_t + \delta_{m,t} + \varepsilon_{i,t}, \quad (1)$$

where i defines the province, m the macro region (the north versus the rest of the country), and t the year. *NewEnterprises* is the logarithm of the number of new enterprises per 100,000 inhabitants. *Mafia* × *Crisis* is an indicator variable for the interaction between our variable for mafia presence, *Mafia*, and an indicator variable *Crisis* for the period starting in 2007. The model also includes the vector $\mathbf{X}_{it}$ with the control variables introduced in section IIIA, province fixed effects (α_i) for time-invariant heterogeneity at the provincial level, year fixed effects (γ_t) for common shocks across provinces, and macro region-year fixed effects ($\delta_{m,t}$) to capture structural differences in the reaction to the crisis across macro regions m. $\varepsilon_{i,t}$ is the error term of the model.

We aim to consistently estimate the coefficient β_1 for *Mafia* × *Crisis*. This parameter captures different trends in the number of new enterprises between provinces with and without mafia after the hit of the subprime mortgage crisis. In general, the crisis negatively hits the number of new enterprises established across the Italian territory (see figure 4). Therefore, positive values of the parameter β_1 would imply a less severe drop in the outcome variable for provinces with mafia presence compared to provinces without mafia presence. On the contrary, a negative parameter would imply a more severe

[12]The after-2006 adjustments in both the demand and supply sides of the cannabis market are likely driven by the 2006 change in the Italian legislation due to Law 49/2006. This law eliminated the distinction between "soft" and "hard" drugs, thus equating cannabis consumption with the consumption of harder drugs.

FIGURE 3.—THE MARKET FOR ILLEGAL DRUGS IN ITALY



This figure shows the evolution over time of the Italian market for illegal drugs. (a) The time trend of the total number of offenses for drug use. (b) The time trend of the total number of seizures. (c) The total number of kilograms seized. (d) The estimated average price. Each panel displays the case of amphetamines, cocaine, heroin, and cannabis. The analysis is based on data provided by the European Monitoring Centre for Drugs and Drug Addiction.

drop in the outcome variable for provinces with mafia than for those without mafia.

In the second specification, we consider an event study setup. The event study allows us to understand the timing of the possible mafia effect and its evolution over time. It takes the following form:

$$NewEnterprises_{i,t} = \beta_0 + \sum_{\tau\in(-4,6),\tau\neq-1} \boldsymbol{\beta}_\tau Mafia_i \times \mathbb{1}(t+\tau) + \\ + \mathbf{X}'_{i,t}\boldsymbol{\beta} + \alpha_i + \gamma_t + \delta_{m,t} + \varepsilon_{i,t}. \qquad (2)$$

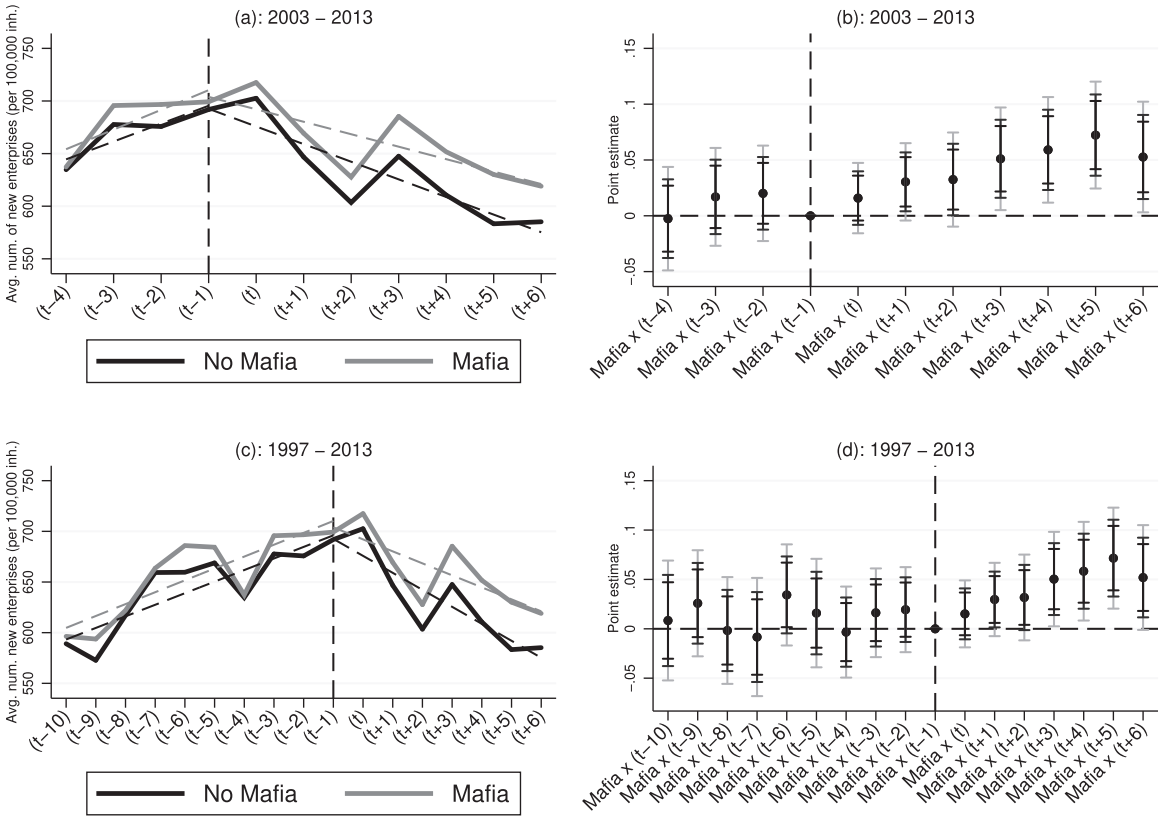
The event study setup takes the same form as equation (1), but instead of pooling different years in a precrisis versus a postcrisis period, we interact *Mafia* with an indicator variable for each year ($t+\tau$). We label as t the year of the subprime mortgage crisis outbreak, 2007. The precrisis period covers the years from $t-4$ to $t-1$, while the postcrisis period covers the years from t to $t+6$. We consider the last precrisis year $t-1$ as the reference year.

The DiD estimation strategy requires the parallel trends assumption to hold, that is, the trends in the number of new enterprises for provinces with and without mafia need to be parallel in the precrisis period. Figure 4 provides a graphical and a more formal test for the parallel trends assumption. The graphical test shows the evolution over time of the number of enterprises between provinces with and without mafia. The second test is based on the event study setup. Specifically, the number of new enterprises established at the province-year level is regressed on the set of interaction terms between *Mafia* and indicator variables for each year.[13] Figures 4a and 4b report the analysis for the period of interest for this study, i.e. 2003 to 2013. As one possible concern of this analysis is that the precrisis period might be unable to capture possible secular trends, in figures 4c and 4d, we replicate the same analysis focusing on a wider precrisis period starting in 1997, the first year with data for the Registry of Enterprises.

The graphical analysis in figures 4a and 4c suggests that the parallel trends assumption holds in our framework. We plot both the trends in the number of new enterprises for provinces with and without mafia presence and the linear fit for such trends in the pre- and postcrisis periods. In the precrisis period, the trend for the number of new enterprises for provinces with no mafia presence fairly mimics the trend for provinces with organized crime. The dotted line for the linear interpolation in the pre- and postcrisis periods suggests that the trends for provinces with and without mafia are parallel and increasing before the crisis, while they start decreasing and

[13]The event study specification only includes province fixed effects, year fixed effects, and macro region-year fixed effects. Section IV describes the analysis with control variables.

FIGURE 4.—TESTING PARALLEL TRENDS



This figure shows the evolution over time of the effect of mafia presence on established enterprises. (a, b) The analysis for 2003–2013. (c, d) The analysis for 1997–2013. Panels a and c report the trends in the number of established enterprises and the linear fit for such trends by mafia presence. The comparison is between provinces within the first two tertiles of the TMI distribution (No Mafia, black line) and those belonging to the third tertile of the distribution (Mafia, gray line). Panels b and d report the event study analysis. The dependent variable is the number of new enterprises (per 100,000 inhabitants, logarithmic scale). The variables *Mafia* $\times$ $(t+\tau)$ are interaction terms of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with indicator variables for each year. Period t identifies the starting year of the subprime mortgage crisis, and the black vertical dashed line visually separates the precrisis and the postcrisis periods. The model only includes province fixed effects, year fixed effects, and macro region-year fixed effects. Panels b and d report each point estimate of *Mafia* $\times$ $(t+\tau)$ for the effect of mafia presence in a specific year together with the 90% (black), 95% (gray), and 99% (light gray) confidence intervals. Confidence intervals are based on spatially (200 km) and serially (11 years) correlated standard errors.

diverging in the postcrisis period. The divergence points to a less severe reduction in the number of postcrisis established enterprises for provinces with mafia than without mafia.

The event study setup in figure 4b confirms the existence of parallel trends. Provinces with and without mafia are on similar trends in the precrisis period. With respect to the last precrisis year $t-1$, none of the three estimated precrisis coefficients is statistically significant at the conventional levels, neither separately nor jointly (p-value $= 0.34$), and point estimates are close to 0. Figure 4d confirms the existence of parallel trends and that when considering a wider time period, any clear secular trend in the number of established enterprises is observed when provinces with and without mafia are compared (joint test p-value $= 0.21$). Also, the event study setup seems to point to a positive effect of mafia presence, namely, a less severe postcrisis drop in the number of established enterprises, for provinces with mafia compared to provinces without mafia.[14]

Given that the parallel trends assumption holds, the DiD identification strategy does not require the group of provinces with and without mafia presence to be balanced in terms of observable characteristics. However, in our framework, the comparison of the control variables between provinces with and without mafia might help clarify whether the two groups of provinces are fairly similar with respect to a set of observable characteristics. Table 1 compares the average precrisis values of our control variables for provinces with versus without mafia presence. Provinces with and without mafia

TABLE 1.—BALANCING TESTS

	(1) Mean No Mafia	(2) Mean Mafia	(3) Difference
Big banks (%)	0.329	0.404	-0.074^{**}
Self-employed (%)	0.285	0.269	0.016^{*}
Tourism index	3.361	2.098	1.264
Wastes per capita (tons)	0.551	0.562	−0.011
Trial duration (days)	3048	3083	−35
Urban population (%)	0.255	0.294	−0.039
Newspaper circulation (1,000 inh.)	122	115	7
Blood donations (bags per 100 inh.)	3.810	3.148	0.662

This table shows the comparison of a set of observable characteristics between provinces without and with mafia presence. The table reports, for each characteristic, the average value over the precrisis period for the group of provinces without mafia presence (first and second tertiles of the TMI distribution) in column 1 and with mafia presence (third tertile of the TMI distribution) in column 2. The sample includes 84 Italian provinces, 56 without mafia presence and 28 with mafia presence. Column 3 reports the difference between columns 1 and 2 and the statistical significance of the difference in means between the two groups of provinces. Significant at *10%, **5%, and ***1%.

[14]We discuss the postcrisis period more in detail in section IV.

TABLE 2.—MAFIA PRESENCE AND NEW ENTERPRISES

	(1) New Enterprises	(2) New Enterprises	(3) New Enterprises	(4) New Enterprises	(5) New Enterprises
Mafia × Crisis	0.038*	0.038**	0.036***	0.039***	0.041***
	(0.020)	(0.017)	(0.013)	(0.013)	(0.011)
Observations	924	924	924	924	924
R^2	0.011	0.017	0.207	0.350	0.409
Province FE	YES	YES	YES	YES	YES
Year FE	NO	YES	YES	YES	YES
North × Years FE	NO	NO	YES	NO	YES
Controls	NO	NO	NO	YES	YES

This table shows the effect of mafia presence on established enterprises in the postcrisis period. The dependent variable is the number of new enterprises (per 100,000 inhabitants, logarithmic scale). The variable *Mafia × Crisis* is the interaction of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with an indicator variable for the postcrisis period. In column 1 we include an indicator variable for the postcrisis period. The set of controls includes variables for the number of big banks, the importance of self-employment, the capacity to attract tourism, waste per capita, trial duration, urban population, newspaper circulation, and blood donations. See section IIIA for further details on control variables. Standard errors are adjusted for spatial (200 km) and serial (11 years) correlation. Significant at *10%, **5%, and ***1%.

presence are fairly similar with respect to control variables, with two variables turning out to be statistically different. Prior to the onset of the crisis, the percentage of big banks is larger in provinces with mafia (0.404 versus 0.329), and the percentage of self-employment is slightly larger in provinces without mafia (0.285 versus 0.269). These two imbalances do not constitute a threat to our identification strategy. First, we include in our specifications control variables for big banks and self-employment. Second, in figure A.10, we report the analysis with the event study setup of control variables that shows that big banks and self-employment in figures A.10a and A.10b are on parallel trends in the precrisis period. All the precrisis coefficients are indeed very small in size and statistically insignificant. Trends also remain parallel in the postcrisis period. Parallel trends in control variables reassure us that there are no differential trends with the potential to affect our baseline estimates.

IV. Baseline Results

Table 2 reports the OLS estimates of equation (1). We estimate five different specifications with the (log) number of new enterprises as the dependent variable. The specification in column 1 includes only province fixed effects. Column 2 includes province and year fixed effects. Column 3 includes province and year fixed effects and macro region-year fixed effects for the north versus the rest of the country. Column 4 includes province and year fixed effects and the set of control variables described in section IIIA. Column 5 includes province and year fixed effects, macro region-year fixed effects, and the set of control variables.[15] We use Conley spatial HAC standard errors (Conley, 1999, 2008) to consider both spatial and serial correlations of mafia presence.[16] We set the spatial lag to 200 km and the temporal lag to eleven years, the entire length of our observational window.[17]

The estimates in table 2 show that provinces with mafia presence experience a less severe postcrisis reduction in the number of newly established enterprises with respect to provinces without mafia. The effect of mafia presence is similar across specifications with improvements in the precision of estimates with the progressive inclusion in the model of control variables. For this reason, in the remainder of the paper, we refer to the most complete specification in column 5 as the "complete model."

Considering the complete model, provinces with mafia presence experience a 4.1% less-severe postcrisis reduction in the number of established enterprises with respect to provinces without mafia. In absolute terms, this effect implies a provincial average of 241 enterprises—28 per 100,000 inhabitants—established every year in the postcrisis period due to mafia investment in the legal economy.

In figure 5, we perform the analysis in the event study setup based on the complete model. First, as for the analysis without control variables in figure 4, provinces with and without mafia are on parallel trends in the precrisis period (joint significance p-value $= 0.46$). Point estimates for the precrisis effect of mafia presence are very close to 0 and never statistically distinguishable from 0. Second, in the postcrisis period a less severe drop—positive coefficients—in the number of established enterprises is observed in provinces with mafia. Point estimates for the effect of mafia presence start to increase over time, producing larger estimates with the worsening of the effect of the crisis. The peak of the effect is reached at year $t + 5$ with an estimated effect above 5%.

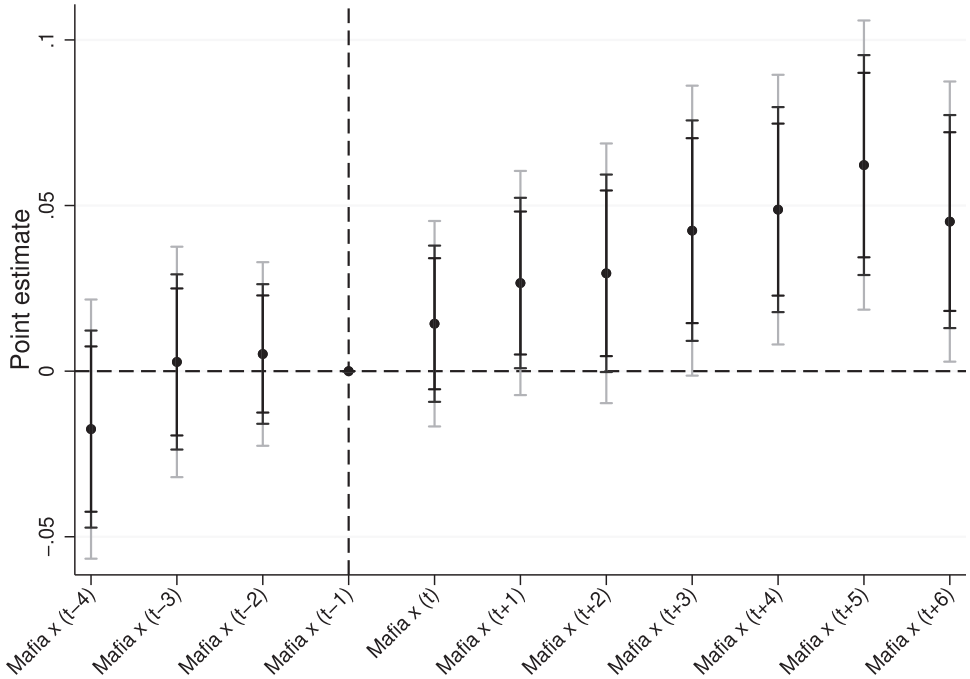
As the definition of mafia is not univocal, in appendix A.4 we replicate the analysis by relaxing each element of the definition of *Mafia* used in the baseline. First, we abstract from the use of the last tertile of the TMI distribution by using different quantiles of the distribution. Second, we relax the

[15]The analysis is robust to the use of more granular definitions for macro region-year fixed effects, for example, northwest, northeast, center, south, and islands.

[16]We compute Conley spatial HAC standard errors with the code developed by Thiemo Fetzer and available at http://www.trfetzer.com/conley-spatial-hac-errors-with-fixed-effects/.

[17]The correlogram analysis in figure A.11 in the appendix suggests the choice of 200 km for the spatial lag. The correlogram shows that the spatial correlation of mafia presence tends to vanish at 200 km from the centroid of the main city in each province.

FIGURE 5.—MAFIA PRESENCE AND NEW ENTERPRISES: EVENT STUDY



This figure shows the evolution over time of the effect of mafia presence on established enterprises. The dependent variable is the number of new enterprises (per 100,000 inhabitants, logarithmic scale). The variables *Mafia* × $(t + \tau)$ are interaction terms of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with indicator variables for each year. Period t identifies the starting year of the subprime mortgage crisis, and the black vertical dashed line visually separates the precrisis and the postcrisis periods. The model is the complete model (see text for further details) and includes province fixed effects, year fixed effects, macro region-year fixed effects, and the set of control variables described in section IIIA. The figure reports each point estimate of *Mafia* × $(t + \tau)$ for the effect of mafia presence in a specific year together with the 90% (black), 95% (gray), and 99% (light gray) confidence intervals. Confidence intervals are based on spatially (200 km) and serially (11 years) correlated standard errors.

use of an indicator variable and rely on a continuous index for mafia presence. Third, we replicate the analysis by using all available measures other than the TMI for mafia presence. All results are consistent with the results of the baseline analysis.

Appendix A.5 provides a set of robustness tests for our baseline results. In particular, we test whether our baseline estimates change in response to the use of different sample selection criteria, to the consideration of possible secular trends in control variables and to the inclusion of additional control variables for financial resources availability at the local level. Finally, in appendix A.6, we use an instrumental variable approach to deal with potential residual endogeneity of mafia presence. All of these tests confirm the robustness of the baseline analysis.

V. Validation, Placebo, and Falsification Tests

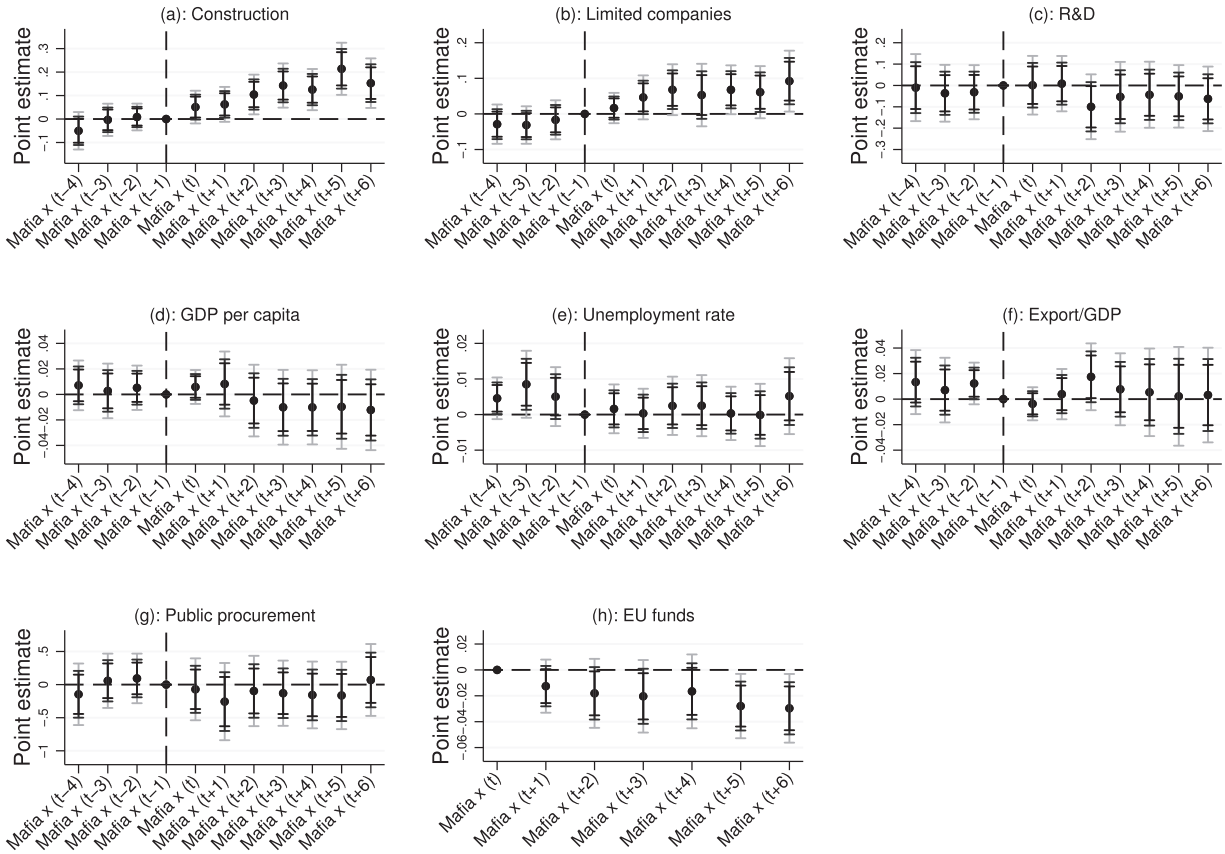
We run validation, placebo, and falsification tests for our baseline results. Validation and placebo tests serve to support the interpretation of the baseline results as evidence of mafia investment in the legal economy. Falsification tests allow us to rule out potential explanations, other than mafia investment, that would drive our baseline results. This section shows the analysis based on the event study setup for the complete model. Table A.6 in the appendix reports the results with the standard DiD approach. The results obtained with the two specifications are similar.

Mafia has strong sectoral and legal form preferences when it comes to investing its money. We would expect to find larger postcrisis mafia effects in sectors or legal forms preferred by mafia, while there should be no or a reduced effect of mafia presence for sectors rarely preferred by mafia. We test this hypothesis by using the number of established enterprises in specific sectors or with a specific legal form as the dependent variable.

The Ministry of the Interior (2013) and Savona (2015) illustrate mafia preferences for its investment in the legal economy. Italian mafia-type organizations are particularly active in the construction sector; around 30% of mafia-type organizations firms seized by Italian authorities operate within this sector. The construction sector is characterized by high movement of capital and high levels of profitability, allowing criminal organizations to launder money. In terms of legal forms, limited companies are by far the first option for criminal organizations (46.6%). Limited companies are particularly easy to establish and require a minimum initial capital of EUR 10,000. Moreover, they guarantee limited liability of business partners. On the contrary, mafia does not heavily invest in the professional, scientific, and technical sector (R&D), as it is highly professionalized and involves a very high level of competition and know-how.

Figures 6a and 6b report the analysis of the construction sector and limited companies, respectively. For both, the analysis shows that provinces with and without mafia are on parallel trends precrisis. After the outbreak of the crisis, provinces

FIGURE 6.—VALIDATION, PLACEBO, AND FALSIFICATION TESTS



This figure shows the validation, placebo, and falsification tests for the baseline analysis. The dependent variables are the number of new enterprises in the construction sector (per 100,000 inhabitants, logarithmic scale, panel a), the number of new enterprises with legal form of limited company (per 100,000 inhabitants, logarithmic scale, panel b), the number of new enterprises in the R&D sector (per 100,000 inhabitants, logarithmic scale, panel c), GDP per capita (panel d), unemployment rate (panel e), exports over GDP (panel f), the amount of public procurement (per 100,000 inhabitants, logarithmic scale, panel g), and the amount of EU funds (percentage over the total amount at the national level, panel h). The variables *Mafia* $\times$ $(t+\tau)$ are interaction terms of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with indicator variables for each year. Period t identifies the starting year of the subprime mortgage crisis, and the black vertical dashed line visually separates the precrisis and the postcrisis periods. Data on EU funds are only available since period t. The model is the complete model (see text for further details) and includes province fixed effects, year fixed effects, macro region-year fixed effects, and the set of control variables described in section IIIA. All figures report each point estimate of *Mafia* $\times$ $(t+\tau)$ for the effect of mafia presence in a specific year together with the 90% (black), 95% (gray), and 99% (light gray) confidence intervals. Confidence intervals are based on spatially (200 km) and serially (11 years) correlated standard errors.

with mafia report a statistically significant less-severe drop with respect to provinces without mafia—positive point estimates for the postcrisis effect of mafia presence—in the number of established enterprises in the construction sector and in the number of established limited companies. Point estimates are in both cases larger than the one in the baseline estimates, confirming mafia preference for the construction sector and for limited companies. Moreover, as in the baseline, the effect becomes larger with the exacerbation of the consequences of the crisis.

Figure 6c depicts the case of the R&D sector. Provinces with and without mafia report parallel trends precrisis. Postcrisis, provinces with and without mafia remain on parallel trends. This evidence reinforces the interpretation of the effect of mafia presence in the baseline as driven by mafia investment in the legal economy. Indeed, while we find large mafia effects in sectors or legal forms usually preferred by mafia, we do not find any mafia effect on sectors not infiltrated by mafia such as R&D.

Five additional falsification tests corroborate the interpretation of our baseline results. The aim of these tests is to confirm that the mafia effect in the baseline analysis is not driven by postcrisis changes in determinants of the establishment of new enterprises that potentially changed in different ways between provinces with and without mafia.

We start by covering demand factors such as GDP per capita and unemployment rate. Differential postcrisis performance in terms of provincial GDP or unemployment might explain the treatment effect of the baseline analysis. In other words, we need to verify that provinces with mafia have not experienced less-severe consequences on the demand for goods induced by the crisis. Otherwise, less-severe consequences of the crisis on the demand for goods in provinces with mafia might explain the less-severe postcrisis decline in the number of established enterprises. Figures 6d and 6e show the analysis of provincial GDP per capita and provincial unemployment rate as dependent variables, respectively. The analysis of GDP suggests that in the postcrisis period,

provinces with and without mafia are on parallel trends. The same occurs for the unemployment rate. The crisis similarly hit provinces with and without mafia in terms of GDP and unemployment. Therefore, such demand factors are unlikely to drive our baseline results.

We then analyze the degree of openness to international markets as measured by the level of exports over GDP. Provinces less connected to the global market might be less affected by the global crisis. If the effect of the crisis was less severe in provinces with mafia presence due to lower connections to the global market, then our estimates might reflect a different impact of the crisis between provinces rather than differences in the level of mafia investment in the legal economy.

We analyze the precrisis degree of openness to international markets as well as its evolution in the postcrisis period. First, in our sample, provinces with and without mafia presence display similar values of exports. The average precrisis values amount to 20% and 22% of GDP for provinces with and without mafia, respectively. The difference between the two values is statistically insignificant. This evidence reassures that our baseline results are unlikely driven by the level of connection to international markets. Second, we test for possible differences in the postcrisis period in the relative importance of exports between provinces with and without mafia presence. If in the postcrisis period exports become relatively less important in provinces with mafia presence than in provinces without mafia presence, this change would contribute to the positive effect of mafia presence in the baseline analysis. Figure 6f shows the analysis with the share of provincial exports with respect to GDP as the dependent variable. With the exception of a weakly significant positive mafia effect in year $t + 2$, all the postcrisis point estimates are close to 0 and statistically insignificant. It is unlikely that factors related to the degree of openness to international markets drive our baseline results.

We run two additional falsification tests to verify whether differential postcrisis changes in financial and economic constraints faced by provinces with and without mafia might explain our baseline results. We focus on the amount of public procurement and EU funds granted to Italian provinces. If the amount of funds granted to provinces with mafia drops less postcrisis than in provinces without mafia, this might drive the mafia effect in the baseline analysis.

We start by analyzing public procurement as a considerable share of mafia's profits comes from bidding for public procurement contracts. The analysis aims to test whether in the postcrisis period, we observe differential changes in the amount of public procurement between provinces with and without mafia. To perform the analysis, we calculate the total amount of public procurement at the provincial level for 2003 to 2013 per 100,000 inhabitants (logarithmic scale).[18] Figure 6g shows the analysis with the amount of public procurement at the province-year level as the dependent variable. In the postcrisis period, provinces with and without mafia display parallel trends in terms of public procurement. Therefore, the analysis of public procurement reassures that the mafia effect in the baseline analysis is unlikely driven by postcrisis changes in public procurement.

As a last falsification test, we analyze EU funds granted to Italian provinces. Provinces with more mafia are usually granted a higher share of EU funds, which are deemed to be less susceptible to the economic cycle. To perform the analysis, we calculate the provincial share of EU funds over the total national amount of EU funds. Unfortunately, data for Italian provinces are only available since 2007.[19] Due to this data limitation, we restrict the event study analysis to the postcrisis period and use t as the reference value for EU funds granted to provinces with and without mafia. Figure 6h reports the postcrisis analysis with EU funds as the dependent variable. There are no sizable differential changes between provinces with and without mafia regarding the amount of EU funds received in the postcrisis period. If anything, mafia provinces experience a slight postcrisis decrease in the amount of EU funds received. This relative decrease poses no concern to the empirical analysis given that if anything, it would be harder to find the positive postcrisis mafia effect documented in the baseline analysis. Despite the restricted sample of years used, the analysis of EU funds suggests that the evolution of postcrisis financial and economic constraints faced by Italian provinces is unlikely to drive our baseline analysis.

VI. The Case of Closed and Registered Enterprises

In this section, we extend our baseline analysis to shed light on the overall mafia effect on local entrepreneurial activity. Mafia presence and investment in the legal economy affect the establishment of new enterprises in addition to the closure of existing enterprises. This study aims to disentangle mafia capital inflows to the legal economy through actual investments from other types of interference in local entrepreneurial activity such as usury. For this reason, we mainly focus on new enterprises. New enterprises are less affected by illegal mafia lending practices. Indeed, although it is impossible to completely rule out the possibility that new entrepreneurs could eventually turn to organized crime to borrow capital for starting a new business, Unioncamere (2014), Sos Impresa (2012), and Dalla Pellegrina (2008) suggest that closed or already existing enterprises are more affected by these practices.

[18]The data on public procurement are elaborated by De Carolis et al. (2019) from records of the National Anticorruption Authority (ANAC). The data contain the universe of public procurement contracts for projects above EUR 40,000. These contracts represent 70% of all public procurement contracts. Data on contracts with a value below EUR 40,000 are limited and only available since 2011.

[19]The data are collected by OpenCoesione. OpenCoesione provides data on all projects financed by European Structural Funds with a national cofinancing requirement.

TABLE 3.—MAFIA PRESENCE, CLOSED, AND REGISTERED ENTERPRISES

	(1) Closed Enterprises	(2) Registered Enterprises
Mafia × Crisis	0.027	0.015*
	(0.018)	(0.009)
Observations	924	924
R^2	0.179	0.510
Province FE	YES	YES
Year FE	YES	YES
North × Years FE	YES	YES
Controls	YES	YES

This table shows the effect of mafia presence on closed and registered enterprises in the postcrisis period. The dependent variables are the number of closed enterprises (per 100,000 inhabitants, logarithmic scale) in column 1, and the number of registered enterprises (per 100,000 inhabitants, logarithmic scale) in column 2. Both dependent variables are expressed as the logarithm of the number of enterprises per 100,000 inhabitants. The variable *Mafia* × *Crisis* is the interaction of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with an indicator variable for the postcrisis period. The set of controls includes variables for the number of big banks, the importance of self-employment, the capacity to attract tourism, waste per capita, trial duration, urban population, newspaper circulation, and blood donations. See section IIIA for further details on control variables. Standard errors are adjusted for spatial (200 km) and serial (11 years) correlation. Significant at *10%, **5%, and ***1%.

We start with the analysis of closed enterprises. The postcrisis effect of mafia presence on closed enterprises is dual. On the one hand, in the postcrisis, mafia presence might induce a negative mafia effect on closures, such as fewer closed enterprises, or last-resort effect, due to mafia investment in the legal economy. Due to worse economic conditions, entrepreneurs might resort to organized crime to obtain resources to run their business more than in the precrisis period. On the other hand, mafia presence and practices, such as usury and racketeering, deteriorate the local arena and the local economy (Reuter et al., 1982; Reuter, 1987, 1993) with consequences that can be exacerbated during recessions. For example, with less disposable income due to the crisis, entrepreneurs are more likely than in the precrisis period to struggle to repay loans taken at usurious rates. This results in a positive postcrisis effect of mafia presence on closures, such as more closed enterprises or disruptive effect. The point estimate for the postcrisis effect of mafia presence clarifies whether the last-resort effect or the disruptive effect prevails in the postcrisis period.

We also analyze registered enterprises, namely, the stock of enterprises operating in a province at a specific point in time. The mafia effect on registered enterprises reveals the combination, or net effect, of the mafia effect on established and closed enterprises. The number of registered enterprises is

$$Registered_{i,t} = Registered_{i,t-1} + (NewEnterprises_{i,t} - Closed_{i,t}), \quad (3)$$

where i is the province and t the year. *NewEnterprises* is the number of new enterprises, and *Closed* is the number of closed enterprises.

Table 3 reports the analysis of closed and registered enterprises based on our complete model. For both closed and registered enterprises, the outcome is expressed as the logarithm of the number of enterprises per 100,000 inhabitants. Column 1 of the table shows the case of closed enterprises. The point estimate for *Mafia* × *Crisis* is positive and statistically insignificant, with an effect that amounts to about 2.7%. Although statistically insignificant and smaller than the effect on established enterprises, the positive sign of the effect on closures might be interpreted as weak evidence of a postcrisis disruptive effect of mafia presence slightly outperforming the last-resort effect.

Column 2 shows the analysis of registered enterprises. The evidence of a positive effect of mafia presence on new enterprises larger than the statistically insignificant increase in closures shapes a positive mafia effect on registered enterprises. The effect amounts to 1.5%, an effect size similar to the difference between the point estimate for the postcrisis mafia effect for established with respect to closed enterprises. In a period in which the stock of operating enterprises tends to decline, registered enterprises operating in provinces with a mafia presence experience a less-severe drop than in provinces without mafia.

The interpretation of the results for closed and registered enterprises is supported by the analysis in the event study setup in figure 7. Figure 7a for closed enterprises reveals parallel precrisis trends in provinces with and without mafia presence. In the postcrisis period, the analysis confirms the dominance of the disruptive effect of mafia presence over its last-resort effect. Indeed, point estimates for the effect of mafia presence become positive although imprecisely estimated.

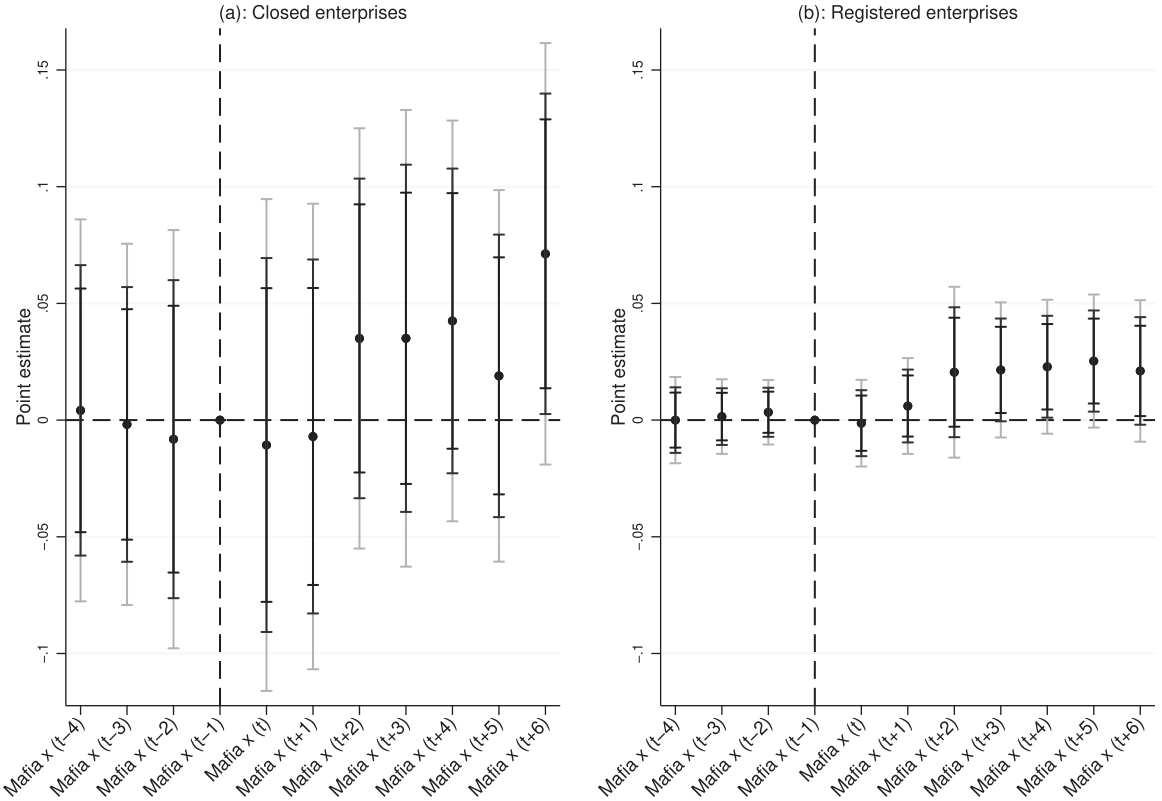
Figure 7b displays the case of registered enterprises. For registered enterprises, the precrisis parallel trends assumption holds. After the outbreak of the subprime mortgage crisis, provinces with mafia presence experience a less severe drop in the stock of operating enterprises than provinces without mafia presence. Point estimates become statistically significant at year $t + 3$.

VII. Conclusion

In this work, we investigate organized crime's investment in the Italian legal economy by exploiting the mandatory registration required by both legal and illegal enterprises. We implement a DiD empirical strategy based on the comparison of new enterprises established before and after the onset of the 2007 subprime mortgage crisis and between provinces with high levels of mafia presence and provinces with low levels of mafia presence.

We find that provinces more infiltrated by organized crime experience a less severe postcrisis drop in established enterprises than less-infiltrated provinces. Quantitatively, in the postcrisis period, the establishment of new enterprises drops by 4% less in more infiltrated provinces than in less-infiltrated provinces. This estimate is likely to represent a lower bound for the actual extent of mafia investment in the legal economy, since it only captures the part of such investment that became detectable due to the crisis. The results of this study are robust to the use of different definitions and measures for

FIGURE 7.—MAFIA PRESENCE, CLOSED, AND REGISTERED ENTERPRISES: EVENT STUDY



This figure shows the evolution over time of the effect of mafia presence on closed and registered enterprises. The dependent variables are the number of closed enterprises (per 100,000 inhabitants, logarithmic scale, panel a) and the number of registered enterprises (per 100,000 inhabitants, logarithmic scale, panel b). The variables *Mafia* $\times$ $(t + \tau)$ are interaction terms of *Mafia*—an indicator variable taking the value of 1 for provinces within the third tertile of the TMI distribution—with indicator variables for each year. Period t identifies the starting year of the subprime mortgage crisis, and the black vertical dashed line visually separates the precrisis and the postcrisis periods. The model is the complete model (see text for further details) and includes province fixed effects, year fixed effects, macro region-year fixed effects, and the set of control variables described in section IIIA. All figures report each point estimate of *Mafia* $\times$ $(t + \tau)$ for the effect of mafia presence in a specific year together with the 90% (black), 95% (gray), and 99% (light gray) confidence intervals. Confidence intervals are based on spatially (200 km) and serially (11 years) correlated standard errors.

mafia presence, to different sample selection criteria, and to the use of different empirical specifications.

Validation and placebo tests based on judicial evidence corroborate the interpretation of the mafia effect on new enterprises as evidence of mafia investment in the legal economy. On the other hand, a set of falsification tests rules out the possibility that other determinants of the establishment of new enterprises different from mafia investment, such as openness to the global market or allocation of funds, drive our results.

The analysis in this paper highlights the desirability of adopting standard repression policies against criminal organizations complemented by institutional interventions, for example, provision of credit or programs to enhance employment opportunities to undermine the roots of the social consensus obtained through mafia investment in the legal economy. Moreover, by showing mafia preferences in terms of sectors and legal forms, we underline where vigilant monitoring and stricter legislation should be improved. Finally, our research suggests that during recessions, vigilance and monitoring should be improved even further to preempt greater future organized crime infiltration in the economy. As Mario Draghi pointed out during his speech to the Italian parliamentary anti-mafia commission in 2009: "During recession firms see their cash flows dry up and watch the market value of their assets fall. Both of these phenomena render companies more easily assailable by organized crime."

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